

CMPSC 480 Midterm Examination

Cumulative assessment of Weeks 1-7

CMPSC 480 - Instructor Solution Deck

Instructor material - do not distribute with the question deck

Solution overview

This instructor deck provides a complete matched solution and grading guidance.

Coverage: Modules A and B: agents, search, adversarial reasoning, MDPs, RL, approximation, and Bayesian inference

Points

100 total

Expected time

110 minutes

Submission

Individual examination PDF

Instructor verification checklist

- ▶ Work independently and show intermediate algebra, frontier state, or factor state.
- ▶ One double-sided handwritten letter-size reference sheet is permitted.
- ▶ A nonprogrammable calculator is permitted; networked devices and communication are prohibited.
- ▶ State every tie-breaking, terminal, reward, and zero-division convention you use.
- ▶ Answers receive credit for justified reasoning; an unsupported final number may receive little credit.
- ▶ Academic integrity rules of the course and institution apply throughout the examination.

Solution 1.a - Agent and search formulation (3 points)

Question focus. Give one concrete element for each PEAS component.

Answer. Performance includes timely collision-free delivery and battery use; environment includes corridors, doors, people, and stations; actuators include motion and lift commands; sensors include localization, range sensing, battery, and task messages.

Solution 1.b - Agent and search formulation (3 points)

Question focus. Define states, actions, transition model, goal test, and path cost for deterministic route planning.

Answer. A state contains position and other path-relevant discrete facts such as carried item or battery bucket; actions are legal moves or task operations; transition applies one action; goal checks delivery completion; cost can combine time and energy with nonnegative units.

Solution 1.c - Agent and search formulation (4 points)

Question focus. Explain why rational behavior does not imply omniscience and name two defensive interface checks.

Answer. Rationality maximizes expected performance using the available percept history and knowledge, so uncertainty can still produce failure. Checks include immutable state keys, legal successor validation, positive costs for UCS/A-star, deterministic ties, and explicit failure.

Solution 2.a - UCS and A-star (5 points)

Question focus. Trace UCS until a goal is removed, including cheaper rediscovery.

Answer. Remove S(0), insert A(4),B(1); remove B(1), improve A to 2 and insert G(9); remove A(2), improve G to 5; remove G(5). Return S-B-A-G with cost 5.

Solution 2.b - UCS and A-star (5 points)

Question focus. Trace A-star priorities and return the selected path.

Answer. After S, A has $f=7$ and B $f=5$; remove B, improved A has $g=2, f=5$ and G has $f=9$; remove A, improve G to $g=f=5$; remove G and return cost 5.

Solution 2.c - UCS and A-star (5 points)

Question focus. Is the heuristic admissible and consistent? Justify from the graph.

Answer. True remaining costs are $h\text{-star}(G)=0$, $h\text{-star}(A)=3$, $h\text{-star}(B)=4$ via A, and $h\text{-star}(S)=5$ via B-A. The listed h equals these values, so it is admissible. Each edge satisfies $h(n) \leq c + h(n\text{-prime})$, including S-B: $5 \leq 1 + 4$ and B-A: $4 \leq 1 + 3$, so it is consistent.

Solution 3.a - Minimax and alpha-beta (5 points)

Question focus. Compute every minimax value and the selected action.

Answer. $\text{MIN}(A)=4$, $\text{MIN}(B)=3$, and $\text{MIN}(C)=2$. MAX chooses A with root value 4.

Solution 3.b - Minimax and alpha-beta (5 points)

Question focus. Trace alpha-beta left to right and identify pruned leaves.

Answer. After A, root $\alpha=4$. At B, first leaf gives $\beta=3$, so $\alpha \geq \beta$ and leaf 8 is pruned. At C, first leaf 5 does not prune because $\beta=5$; second leaf gives 2. The root still chooses A.

Solution 3.c - Minimax and alpha-beta (5 points)

Question focus. Design a depth-cutoff evaluation with three features and state two tests.

Answer. A valid MAX-perspective weighted sum might use score difference, mobility difference, and threat or distance advantage with normalized scales. Test terminal states bypass the heuristic and alpha-beta returns the same value as plain minimax on generated small trees.

Solution 4.a - MDP planning (5 points)

Question focus. Compute $Q(s,a)$ under reward-on-transition semantics.

Answer. $Q(s,a)=0.8[5+0]+0.2[0+0.9(2)]=4+0.36=4.36$.

Solution 4.b - MDP planning (5 points)

Question focus. Compute $Q(s,b)$ and the greedy action.

Answer. $Q(s,b)=3$ because the transition terminates and has no bootstrap. Since $4.36 > 3$, choose a.

Solution 4.c - MDP planning (5 points)

Question focus. Explain why value iteration converges for a finite discounted MDP.

Answer. The Bellman optimality operator changes values by at most γ times the previous max-norm difference: expectation and maximum do not enlarge the bound beyond γ . Since $\gamma < 1$, it is a contraction with a unique fixed point.

Solution 5.a - Q-learning and approximation (5 points)

Question focus. Compute current Q , target, TD error, and new weights.

Answer. Current $Q=0.5-1=-0.5$. Target $=2+0.5(3)=3.5$; $\text{delta}=4$.
Increment $=0.2(4)[1,2]=[0.8,1.6]$, so new $w=[1.3,1.1]$.

Solution 5.b - Q-learning and approximation (5 points)

Question focus. Recompute the update if the transition is terminal.

Answer. Target=2, $\text{delta}=2-(-0.5)=2.5$, $\text{increment}=0.2(2.5)[1,2]=[0.5,1]$, and new $w=[1,0.5]$.

Solution 5.c - Q-learning and approximation (5 points)

Question focus. Name the deadly triad and two observable instability signals.

Answer. The triad is function approximation, bootstrapping, and off-policy learning. Signals include exploding weight or Q norms, increasing absolute TD-error quantiles, nonfinite values, oscillatory loss or return, and large cross-seed variance.

Solution 6.a - Bayesian networks and inference (5 points)

Question focus. Compute $P(R,S,W)$ for all three true.

Answer. Use the factorization $P(R)P(S|R)P(W|R,S)=0.2(0.1)(0.99)=0.0198$.

Solution 6.b - Bayesian networks and inference (5 points)

Question focus. Write an exact expression for $P(R|W)$ without evaluating missing CPT rows.

Answer. $P(R|W)$ is proportional to $P(R) \sum_s P(s|R)P(W|R,s)$; compute that mass for $R=\text{true}$ and false using their CPT rows, then normalize the two masses.

Solution 6.c - Bayesian networks and inference (5 points)

Question focus. If factors $f(R,S)$ and $g(S,W)$ are multiplied and S is summed out, what is the result scope and why can order matter?

Answer. The product scope is $\{R,S,W\}$; summing S yields a factor over $\{R,W\}$. Different elimination orders can create intermediate factors over more variables and therefore require exponentially more space and arithmetic.

Solution 7.a - Integrated engineering scenario (5 points)

Question focus. Specify the belief-to-policy interface and three invariants.

Answer. Pass a fixed-order normalized finite belief vector, optionally with observable state features, never the hidden truth. Invariants include correct dimension, nonnegative entries summing to one, and legal-action-only output.

Solution 7.b - Integrated engineering scenario (5 points)

Question focus. Design a controlled ablation showing whether belief tracking helps.

Answer. Compare the full agent with a fixed-prior or observation-only agent using identical training steps, seeds, environment cases, and epsilon-zero evaluation; report return, success, hazard rate, and variability.

Solution 7.c - Integrated engineering scenario (5 points)

Question focus. State a response to impossible evidence and a posttraining safety gate.

Answer. Raise or apply a declared fallback posterior when normalization is zero. Before acceptance, require belief oracle tests, terminal-update tests, finite Q /weight limits, and hazard-rate performance across noise levels.

Edge-case reasoning and expected behavior

- ▶ Return an empty path.
- ▶ Use stable action order.
- ▶ Ignore the bootstrap.
- ▶ Raise or use the declared fallback.
- ▶ Choose a lower-growth elimination order and report maximum factor scope.

Grading rubric - part 1

Category	Points	Full-credit evidence
Agent and search formulation	10	Components are precise and connected to observable behavior.
UCS and A-star	15	Costs, updates, and guarantee conditions are correct.
Minimax and alpha-beta	15	Values, bounds, pruning, and perspective are correct.
MDP planning	15	Expectations, terminal convention, and contraction logic are sound.

Grading rubric - part 2

Category	Points	Full-credit evidence
Q-learning and approximation	15	Arithmetic and deadly-triad reasoning are precise.
Bayesian networks and inference	15	CPT selection, normalization plan, and factor scope are correct.
Integrated engineering scenario	15	The answer connects inference, learning, evaluation, and defensive controls.

Reflection exemplar

Answer. A full-credit response names the problem, states the assumption, and explains the exact trace, equation, or guarantee affected.

Grading guidance. Award credit for a specific claim supported by technical evidence from the submitted work.